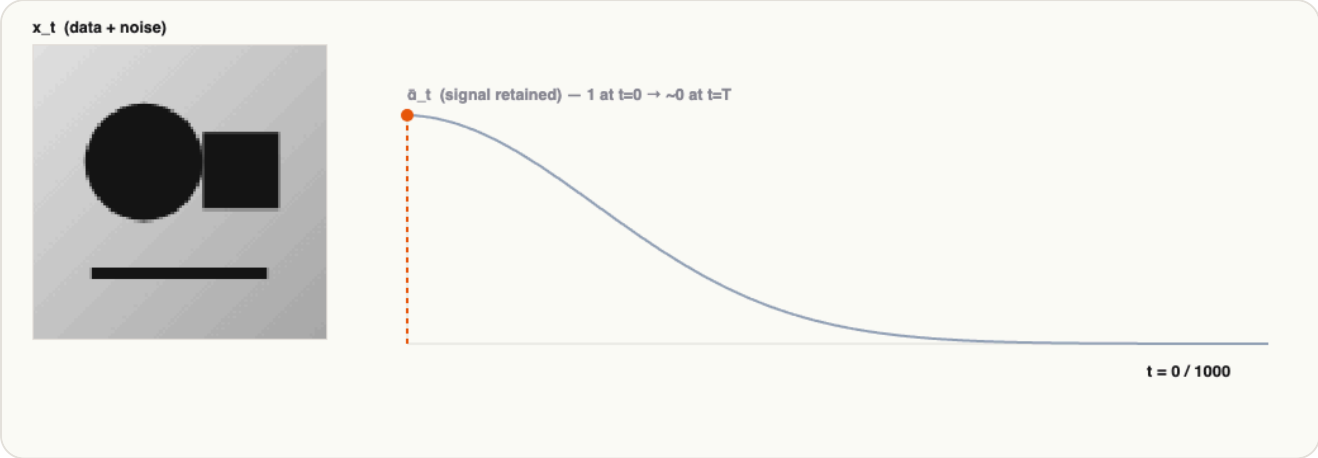
Diffusion Models — Cheat Sheet & Visualizer

DDPM → ELBO → denoising → U-Net → score → guidance → LDM → DDIM

Hierarchical VAEs with a fixed encoder: the full DDPM derivation, score equivalence, and DDIM.

Interactive — forward noising & reverse (DDIM, σ=0)

The forward process is fixed, not learned: $x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\,\epsilon_t$. Scrub t to destroy the image, then run the reverse. With an *oracle* denoiser and $\sigma_t = 0$, the reverse is fully deterministic — exactly the DDIM update on page 110.



`x_t = √ᾱ_t·x₀ + √(1−ᾱ_t)·ε  ·  ᾱ_t = 1.00000  ·  signal √ᾱ_t = 1.000  ·  noise √(1−ᾱ_t) = 0.000`

p72 Denoising Diffusion probabilistic Models (DDPMs)

$$\text{Given Data } D = \{x_0\} \sim P_{x_0} \quad \text{(data distribution)}$$

Goal: To learn to sample from P_{x_0} (Gen.AI)

DDPMs are latent variable Generative Models seen as **special case of hierarchical VAEs**.

Recall, in a VAE:

$$\underbrace{x}_{\text{Data}} \xrightarrow[q(z|x)]{\text{Enc}} \underbrace{z}_{\text{latent}} \xrightarrow[P(x|z)]{\text{Dec}} \underbrace{x}_{\text{Data}}$$

In a **HVAE**, there are **multiple latent spaces**.

$$x \to z_1 \to z_2 \to z_3 \to \cdots \to z_T \qquad \text{(Encoding)}$$

$$z_T \to z_{T-1} \to z_{T-2} \to \cdots \to x \qquad \text{(Decoding)}$$

A DDPM is a h-VAE, with the following properties: **(a)** There are multiple latent spaces: $z_1, z_2, \ldots, z_T$

p73 DDPM vs VAE; notation

(b) The dimensionality of the latent spaces is **same as that of the data space**.

$$\dim(z_t) = \dim(x) \quad \forall t$$

(c) The **Encoding procedure is fixed or non-learnable**, unlike a VAE.

$$\text{VAE} : q_\varphi(z|x) \quad : \varphi \text{ is a NN that learned}$$

$$\text{DDPM} : q(z|x) \quad : \text{is fixed \& not learnable}$$

$$\text{VAE} : \text{both Encoding \& Decoding are learned}$$

$$\text{DDPM} : \text{only decoding is learned}$$

DDPM Formulation — Notations

- Data variable is denoted by x_0 (x)
- Latent space : $x_1, x_2, \ldots, x_T$ $(z_1, z_2, \ldots, z_T)$
- (not to be confused with previous notations where $x_1, x_2, \ldots$ were data points)
- In the DDPM literature, x_0 : data point; $x_1, x_2, \ldots, x_T$: latent spaces / vectors.

p74 Encoding / Forward process

$$x_0 \to x_1 \to x_2 \to \cdots \to x_T$$

x_0: data point; $x_1 \ldots x_T$: latent vectors corresponding to x_0

$$\dim(x_t) = \dim(x_0) \doteq \mathbb{R}^d \qquad \forall t$$

T : hyperparameter

Define the Encoding / Forward process in DDPM

$$x_1 = \sqrt{\alpha_1}\, x_0 + \sqrt{1-\alpha_1}\, \epsilon_1, \qquad \epsilon_1 \sim \mathcal{N}(0, I)$$

$$x_2 = \sqrt{\alpha_2}\, x_1 + \sqrt{1-\alpha_2}\, \epsilon_2, \qquad \epsilon_2 \sim \mathcal{N}(0, I)$$

$$\vdots$$

$$\boxed{x_t = \sqrt{\alpha_t}\, x_{t-1} + \sqrt{1-\alpha_t}\, \epsilon_t, \qquad \epsilon_t \sim \mathcal{N}(0, I)}$$

where $\alpha_1 \ldots \alpha_T$ are fixed scalars $[0, 1]$.

DIAGRAM $\sqrt{(\alpha_1)}x_0$ (image) + $\sqrt{(1-\alpha_1)}\varepsilon_1$ (noise) → $\sqrt{(\alpha_2)}x_1$ (image) + $\sqrt{(1-\alpha_2)}\varepsilon_2$ (noise) → ⋯ → x_T

$$\underbrace{x_0}_{\text{Data}} \to \text{add noise} \to \underbrace{x_1}_{\text{noisy data}} \to \text{add noise} \to \underbrace{x_2}_{\text{noisy-noisy data}} \cdots \to \underbrace{x_T}_{\text{"noise only"}}$$

p75 Markov chain and forward conditionals

$$\dot{x}_t = \sqrt{\alpha_t} \cdot x_{t-1} + \sqrt{1-\alpha_t} \cdot \epsilon_t \qquad \epsilon_t \sim \mathcal{N}(0, I) \qquad -(1)$$

The above defines the **Encoding** or the **Forward process** of a DDPM. Eq (1) : represents a **1st order markov chain**.

1st order Markov Chain defines a series of RVs, $x_0, x_1, \ldots, x_T$ s.t.

$$x_t \perp\!\!\!\perp x_{t-2} \ldots x_0 \mid x_{t-1}$$

$\Rightarrow$ One can define a series of Conditional distributions on the latent variables as:

$$q(x_t \mid x_{t-1}) = \mathcal{N}\left(x_t;\ \underbrace{\sqrt{\alpha_t} \cdot x_{t-1}}_{\text{mean}},\ \underbrace{(1-\alpha_t)I}_{\text{var}}\right)$$

$$\text{because, } \dot{x}_t = \sqrt{\alpha_t} \cdot x_{t-1} + \sqrt{1-\alpha_t} \cdot \epsilon_t, \qquad \epsilon_t \sim \mathcal{N}(0, I)$$

- $q(x_t|x_{t-1})$: **Conditional forward / Encoding distributions.**
- $q(x_T) \sim \mathcal{N}(0, I)$: **Stationary distribution of the Markov Chain.**

p76 Model via a decoding distribution

$$P_\theta \left(\underbrace{x_0}_{\text{data}}, \underbrace{x_1, x_2, \ldots, x_T}_{\text{lat. var}} \right) = P_\theta(x_T) \prod_{t=1}^{T} P_\theta \left(x_{t-1} \mid x_t \right)$$

$$\text{where } P_\theta \left(x_{t-1} \mid x_t \right) \doteq \mathcal{N}(x_{t-1};\ \mu_\theta(x_t),\ \Sigma_\theta(x_t))$$

P_θ defines a latent variable model with multiple latent variables, similar to VAE, where we define $P_\theta(x, z)$: joint dist. between data & lat. var.

The definition of the model distribution $P_\theta(\cdot)$ in a DDPM, arises because of the **Markovian (1st order) assumption in the reverse direction**, with **Gaussian transitions containing learnable parameters**.

$$P_\theta \left(x_{t-1} \mid x_t \right) \doteq \mathcal{N}(x_{t-1};\ \mu_\theta(x_t),\ \Sigma_\theta(x_t))$$

$$\mu_\theta(x_t) \ \& \ \Sigma_\theta(x_t) \text{ are learned via ELBO.}$$

p77 ELBO for the DDPM

Recall, in a VAE:

$$\log P_\theta(x) = \log \int_z P_\theta(x, z)\, dz, \qquad J_\theta(q_\varphi) = \mathbb{E}_{q_\varphi(z|x)} \log \frac{P_\theta(x, z)}{q_\varphi(z|x)} \quad : \text{ Evidence lower Bound (ELBO)}$$

In a DDPM,

$$J_\theta(q)^{DDPM} = \mathbb{E}_{q(x_1, x_2..x_T|x_0)} \log \frac{P_\theta(x_0, x_1, x_2.., x_T)}{q(x_1, x_2..x_T|x_0)}$$

$$\boxed{J_\theta(q)^{DDPM} = \mathbb{E}_{q(x_{1:T}|x_0)}\ \log \frac{P_\theta(x_{0:T})}{q(x_{1:T}|x_0)}}$$

$$x_{1:T} = (x_1, x_2, \ldots, x_T), \qquad x_{0:T} = (x_0, x_1, \ldots, x_T)$$

p78 Expanding numerator and denominator

We have,

$$P_\theta(x_{0:T}) = P(x_T) \prod_{t=1}^{T} P_\theta \left(x_{t-1} \mid x_t \right), \qquad P(x_T) = \mathcal{N}(0, I)$$

Similarly,

$$q \left(x_{1:T} \mid x_0 \right) = \prod_{t=1}^{T} q \left(x_t \mid x_{t-1} \right)$$

$$q(x_{1:T}|x_0) = q(x_1|x_0) \cdot q(x_2|x_1, x_0) \cdot q(x_3|x_2, x_1, x_0) \ldots q(x_T|x_{T-1}, x_{T-2} \ldots x_0)$$

$$= \prod_{t=1}^{T} q(x_t|x_{t-1}) \qquad \because \text{ the forward / Encoding is first order Markov}$$

p79 Splitting off the t=1 term

$$\therefore\ J_\theta(q) = \mathbb{E}_q \log \frac{P_\theta(x_{0:T})}{q(x_{1:T}|x_0)} = \mathbb{E}_q \log \left[\frac{P(x_T) \cdot \prod_{t=1}^{T} P_\theta(x_{t-1}|x_t)}{\prod_{t=1}^{T} q(x_t|x_{t-1})} \right]$$

Consider

$$\log \left[\frac{P(x_T) \cdot \prod_{t=1}^{T} P_\theta(x_{t-1}|x_t)}{\prod_{t=1}^{T} q(x_t|x_{t-1})} \right]$$

$$= \log \left[\frac{P(x_T) \cdot P_\theta(x_0|x_1) \cdot \prod_{t=2}^{T} P_\theta(x_{t-1}|x_t)}{q(x_1|x_0) \cdot \prod_{t=2}^{T} q(x_t|x_{t-1})} \right]$$

$$= \log \frac{P(x_T) P_\theta(x_0|x_1)}{q(x_1|x_0)} + \log \prod_{t=2}^{T} \frac{P_\theta(x_{t-1}|x_t)}{q(x_t|x_{t-1})}$$

p80 Bayes rule on the denominator (3 RVs)

Consider the denominator in the second term

$$q(x_t|x_{t-1}) = q(x_t|x_{t-1}, x_0) \qquad \because \text{ 1st order Markov}$$

$$= \frac{q(x_{t-1}|x_t, x_0) \cdot q(x_t|x_0)}{q(x_{t-1}|x_0)} \qquad : \text{ Baye's rule with 3 Rvs.}$$

$q\left(x_{t-1} \mid x_t, x_0\right)$: dist. of x_{t-1}, given that the Encoding process has landed at x_t at $t = t$, starting from x_0.

Plugging the above expression on to the ELBO,

$$\log \frac{P(x_T) P_\theta(x_0|x_1)}{q(x_1|x_0)} + \log \prod_{t=2}^{T} \frac{P_\theta(x_{t-1}|x_t)}{q(x_t|x_{t-1})}$$

$$= \log \frac{P(x_T) P_\theta(x_0|x_1)}{q(x_1|x_0)} + \log \prod_{t=2}^{T} \left(\frac{P_\theta(x_{t-1}|x_t)}{\frac{q(x_{t-1}|x_t, x_0)\, q(x_t|x_0)}{q(x_{t-1}|x_0)}} \right)$$

$$= ____ + \log \prod_{t=2}^{T} \left(\frac{P_\theta(x_{t-1}|x_t)\, q(x_{t-1}|x_0)}{q(x_{t-1}|x_t, x_0) \cdot q(x_t|x_0)} \right)$$

p81 Telescoping product → the final per-term form

$$= \log \frac{P(x_T) P_\theta(x_0|x_1)}{q(x_1|x_0)} + \log \prod_{t=2}^{T} \frac{P_\theta(x_{t-1}|x_t)}{q(x_{t-1}|x_t, x_0)} + \left(\log \prod_{t=2}^{T} \frac{q(x_{t-1}|x_0)}{q(x_t|x_0)} \right)$$

$$\text{Note: } \log \prod_{t=2}^{T} \frac{q(x_{t-1}|x_0)}{q(x_t|x_0)} = \log \frac{q(x_1|x_0) \cancel{q(x_2|x_0)}}{\cancel{q(x_2|x_0)} \cancel{q(x_3|x_0)} \cdots} \text{ (telescopes)}$$

$$= \log \frac{P(x_T) P_\theta(x_0|x_1)}{q(x_1|x_0)} + \log \prod_{t=2}^{T} \frac{P_\theta(x_{t-1}|x_t)}{q(x_{t-1}|x_t, x_0)} + \log \frac{q(x_1|x_0)}{q(x_T|x_0)}$$

$$= \log P_\theta(x_0|x_1) + \log \frac{P(x_T)}{q(x_T|x_0)} + \log \prod_{t=2}^{T} \frac{P_\theta(x_{t-1}|x_t)}{q(x_{t-1}|x_t, x_0)}$$

$$= \log P_\theta(x_0|x_1) + \log \frac{P(x_T)}{q(x_T|x_0)} + \sum_{t=2}^{T} \log \frac{P_\theta(x_{t-1}|x_t)}{q(x_{t-1}|x_t, x_0)}$$

$$\text{the } \sum \text{ block is labelled "Term"}$$

Bringing in the outer expectation term,

p82 Applying the outer expectation correctly

$$\mathbb{E}_{q(x_{1:T}|x_0)}\Big(\text{Term}\Big)$$

Since the sub-terms in (Term) depends only on **subsets** of $x_{1:T}$, the outer expectation can be written appropriately.

$$J_\theta(q) = \mathbb{E}_{q(x_1|x_0)} \log P_\theta(x_0|x_1) + \mathbb{E}_{q(x_T|x_0)} \log \frac{P(x_T)}{q(x_T|x_0)}$$

$$+ \sum_{t=2}^{T} \mathbb{E}_{q(x_{t-1}, x_t|x_0)} \log \frac{P_\theta(x_{t-1}|x_t)}{q(x_{t-1}|x_t, x_0)}$$

$$\text{Consider } \mathbb{E}_{(x_{t-1}, x_t|x_0)}(\,\cdot\,) = \mathbb{E}_{q(x_t|x_0)} \cdot \mathbb{E}_{q(x_{t-1}|x_t, x_0)}(\,\cdot\,)$$

$$\text{property of the cond. expectation}$$

Then: *Rewriting the ELBO using the property of cond. exp.*

p83 The three terms of the DDPM ELBO

$$J_\theta(q) = \mathbb{E}_{q(x_1|x_0)} \log P_\theta(x_0|x_1) + \mathbb{E}_{q(x_T|x_0)} \log \frac{P(x_T)}{q(x_T|x_0)}$$

$$+ \sum_{t=2}^{T} \mathbb{E}_{q(x_t|x_0)} \cdot \mathbb{E}_{q(x_{t-1}|x_t, x_0)} \left(\log \frac{P_\theta(x_{t-1}|x_t)}{q(x_{t-1}|x_t, x_0)} \right)$$

$$\boxed{\begin{aligned} J_\theta(q) = \mathbb{E}_{q(x_1|x_0)} \log P_\theta(x_0|x_1) - D_{KL}\Big(q(x_T|x_0) \,\Big\|\, P(x_T)\Big) \\ - \sum_{t=2}^{T} \mathbb{E}_{q(x_t|x_0)} \Big[D_{KL}\Big(q(x_{t-1}|x_t, x_0) \,\Big\|\, P_\theta(x_{t-1}|x_t)\Big)\Big] \end{aligned}}$$

The above objective of DDPM contains three terms: 1. $\mathbb{E}_{q(x_1|x_0)} \log P_\theta(x_0|x_1)$: **Reconstruction term** 2. $D_{KL}\left(q(x_T|x_0) \,\|\, P(x_T)\right)$: **prior matching term** — *Independent of* θ 3. $\sum_{t=2}^{T} \mathbb{E}_{q(x_t|x_0)} \left[D_{KL}\left(q(x_{t-1}|x_t, x_0) \,\|\, P_\theta(x_{t-1}|x_t)\right)\right]$ - $q(x_{t-1}|x_t, x_0)$: **"Known" denoising** - $P_\theta(x_{t-1}|x_t)$: **learnable denoising**

p84 Consistency (denoising matching) term

Consistency or denoising matching term.

Objective: $\theta^* = \arg\max_\theta J_\theta(q)$ (ELBO)

Estimating the D_{KL} in the Consistency term

$$D_{KL}\Big(q(x_{t-1}|x_t, x_0) \,\Big\|\, P_\theta(x_{t-1}|x_t)\Big)$$

To compute $q(x_{t-1}|x_t, x_0)$

$$q(x_{t-1}|x_t, x_0) = \frac{q(x_t|x_{t-1}, x_0) \cdot q(x_{t-1}|x_0)}{q(x_t|x_0)} \qquad : \text{Baye's rule}$$

$$= \frac{q(x_t|x_{t-1}) \cdot q(x_{t-1}|x_0)}{q(x_t|x_0)}$$

We know

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t;\ \sqrt{\alpha_t} \cdot x_{t-1},\ (1-\alpha_t)I)$$

p85 Recursion to obtain q(x_t | x_0)

$q(x_t|x_0)$ can be obtained using **recursion** as below:

We have,

$$x_t = \sqrt{\alpha_t}\, x_{t-1} + \sqrt{1-\alpha_t}\, \epsilon_{t-1}, \qquad \epsilon_{t-1} \sim \mathcal{N}(0, I)$$

$$= \sqrt{\alpha_t} \left(\sqrt{\alpha_{t-1}}\, x_{t-2} + \sqrt{1-\alpha_{t-1}}\, \epsilon_{t-2} \right) + \sqrt{1-\alpha_t}\, \epsilon_{t-1}, \qquad \epsilon_{t-2} \sim \mathcal{N}(0, I)$$

$$= \sqrt{\alpha_t \alpha_{t-1}}\, x_{t-2} + \underbrace{\sqrt{\alpha_t - \alpha_t \alpha_{t-1}}\, \epsilon_{t-2} + \sqrt{1-\alpha_t}\, \epsilon_{t-1}}_{T_2}$$

$$\text{consider } T_2 = \sqrt{\alpha_t - \alpha_t \alpha_{t-1}}\, \epsilon_{t-2} + \sqrt{1-\alpha_t}\, \epsilon_{t-1} = a \cdot \epsilon_1 + b \cdot \epsilon_2, \qquad \epsilon_1, \epsilon_2 \sim \text{iid}\ \mathcal{N}(0, I)$$

T_2 is a **linear combination of two normal RVs** \& $\therefore$ it'll lead to another Gausn. Rv.

$$T_2 = \epsilon^* \sim \mathcal{N}(0,\ (1 - \alpha_t \alpha_{t-1})I)$$

$$x_t = \sqrt{\alpha_t \alpha_{t-1}}\, x_{t-2} + \left(\sqrt{1 - \alpha_t \alpha_{t-1}} \right) \epsilon^*_{t-2}$$

continuing the recursion till x_0, $\epsilon^*_{t-2} \sim \mathcal{N}(0, I)$

p86 Closed form and the consistency-term ingredients

$$x_t = \sqrt{\alpha_t \alpha_{t-1} \dots \alpha_1}\, x_0 + \sqrt{1 - \alpha_t \alpha_{t-1} \dots \alpha_1}\, \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I)$$

$$x_t = \sqrt{\bar{\alpha}_t} \cdot x_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon$$

$$\text{where } \bar{\alpha}_t = \prod_{i=1}^{t} \alpha_i, \qquad \epsilon \sim \mathcal{N}(0, I)$$

$$q(x_t \mid x_0) = \mathcal{N}\big(x_t;\ \sqrt{\bar{\alpha}_t} \cdot x_0,\ (1 - \bar{\alpha}_t)I\big)$$

$\Rightarrow$ the tth latent can be obtained directly from the data (x_0), without hopping for t-steps.

Getting back to the consistency term

$$q(x_{t-1} \mid x_t, x_0) = \frac{q(x_t \mid x_{t-1}) \cdot q(x_{t-1} \mid x_0)}{q(x_t \mid x_0)}$$

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{\alpha_t}\, x_{t-1},\ (1-\alpha_t)I)$$

$$q(x_t|x_0) = \mathcal{N}\big(x_t; \sqrt{\bar{\alpha}_t}\, x_0,\ (1-\bar{\alpha}_t)I\big)$$

$$q(x_{t-1}|x_0) = \mathcal{N}\big(x_{t-1}; \sqrt{\bar{\alpha}_{t-1}} \cdot x_0,\ (1-\bar{\alpha}_{t-1})I\big)$$

p87 Closed form for q(x_{t-1} | x_t, x_0)

$$\Rightarrow\ q(x_{t-1}\mid x_t,x_0)\propto\frac{\exp\left[-\|x_t-(\cdot)\|_2^2\right]\exp\left(-\|x_{t-1}-(\cdot)\|_2^2\right)}{\exp[-\|x_{t-1}-(\cdot)\|_2^2]}\ \propto\ \exp\left[-\|x_{t-1}-\mu_q\|_2^2\right]$$

obtained via completion of the square method

$$q(x_{t-1}\mid x_t,x_0)=\mathcal{N}(x_{t-1};\ \mu_q(x_t,x_0),\ \Sigma_q)$$

where

$$\mu_q(x_t,x_0)=\frac{\sqrt{\alpha_t}\,(1-\bar\alpha_{t-1})\cdot x_t\ +\ \sqrt{\bar\alpha_{t-1}}\,(1-\alpha_t)\cdot x_0}{1-\bar\alpha_t}$$

$$\Sigma_q=\underbrace{\frac{(1-\alpha_t)(1-\bar\alpha_{t-1})}{1-\bar\alpha_t}}_{\sigma_q^2}\cdot I=\sigma_q^2\cdot I$$

Both μ_q & Σ_q and $\therefore q(x_{t-1}|x_t,x_0)$ are **computable & known**.

p88 The KL becomes a regression on μ_q

$$D_{KL}\left(q(x_{t-1}|x_t,x_0)\,\middle\|\,P_\theta(x_{t-1}|x_t)\right)$$

$$P_\theta(x_{t-1}|x_t)=\mathcal{N}(x_{t-1};\ \mu_\theta,\ \Sigma_q)$$

$$\Rightarrow\ D_{KL}\left(q\parallel P_\theta\right)=D_{KL}\left[\mathcal{N}(x_{t-1};\mu_q,\Sigma_q)\,\middle\|\,\mathcal{N}(x_{t-1};\mu_\theta,\Sigma_q)\right]$$

$$=\frac{1}{2}\left[\log\frac{|\Sigma_q|}{|\Sigma_q|}-d+\mathrm{tr}\Big(\underbrace{\Sigma_q^{-1}\Sigma_q}_{I}\Big)+(\mu_\theta-\mu_q)^T\ \underbrace{\Sigma_q^{-1}}_{(\sigma_q^2 I)^{-1}}\ (\mu_q-\mu_\theta)\right]$$

$$=\frac{1}{2\sigma_q^2}\|\mu_\theta-\mu_q\|_2^2\qquad:\ \text{Regression over }\mu_q$$

p89 ELBO for DDPM (boxed) + the reconstruction term

$$\boxed{J_\theta(q)=\mathbb{E}_{q(x_1|x_0)}\log P_\theta(x_0|x_1)-\sum_{t=2}^{T}\mathbb{E}_{q(x_t|x_0)}\frac{1}{2\sigma_q^2}\|\mu_\theta-\mu_q\|_2^2}\qquad\text{where } x_T\sim\mathcal{N}(0,I)$$

To compute the reconstruction term:

$$\text{Consider }\ \log P_\theta(x_0|x_1)$$

We know $P_\theta(x_{t-1}|x_t)=\mathcal{N}(x_{t-1};\mu_\theta(x_t),\Sigma_q)$

$$\Rightarrow P_\theta(x_0|x_1)=\mathcal{N}\left(x_0;\ \mu_\theta(x_1),\ \sigma_q^2 I\right)$$

$$\log P_\theta(x_0|x_1)=\log\frac{\exp}{(2\pi\sigma_q^2)^{d/2}}-\left\{\frac{\|x_0-\mu_\theta(x_1)\|_2^2}{2\sigma_q^2}\right\}$$

$$=-\frac{\|x_0-\mu_\theta(x_1)\|_2^2}{2\sigma_q^2}-\left(\frac{d}{2}\log 2\pi\sigma_q^2\right)$$

the second term is labelled "Ind of θ"

$$\Rightarrow\log P_\theta(x_0|x_1)\propto-\frac{\|x_0-\mu_\theta(x_1)\|^2}{2\sigma_q^2}$$

p90 Final objective + the U-net approximator

$$J_\theta(q) = \mathbb{E}_{q(x_1|x_0)}\left(-\|x_0 - \mu_\theta(x_1)\|_2^2\right) + \left(-\sum_{t=2}^{T} \mathbb{E}_{q(x_t|x_0)}\left[\frac{1}{2\sigma_q^2}\|\mu_\theta(x_t) - \mu_q\|_2^2\right]\right)$$

$$\theta^* = \arg\max_\theta J_\theta(q)$$

Basic idea is to express the **unknown** $\mu_\theta(x_t)$ using a **function approximator (NN)**.

> **DIAGRAM** x_t, t → [U-net θ (decrease dim → increase dim)] → $\mu_\theta(x_t)$ → $\|\mu_\theta - \mu_q\|_2^2$ (Regressor on μ_q)

- One NN which predicts $\mu_\theta(x_t)$ for any x_t.
- To ensure that NN has an idea of **"time"**, another fixed variable t, corresponding to the "time" of x_t is given as input.
- t is converted into a vector using **sinusoidal positional Embeddings**.

p91 Positional embeddings + ELBO equivalence

positional Embedding: $p : t \to \hat{t},\ t \in \mathbb{Z}^+,\ \hat{t} \in \mathbb{R}^d$

$$\hat{t}(t)_i = \begin{cases} \sin(\omega_k t) & \text{if } i = 2k \\ \cos(\omega_k t) & \text{if } i = 2k+1 \end{cases}$$

$$\omega_k = \frac{1}{(10000)^{2k/d}}$$

ELBO Equivalence:

$$\text{Recall, } \frac{1}{2\sigma_q^2}\|\mu_\theta(x_t) - \mu_q(x_t, x_0)\|_2^2$$

$$\mu_q(x_t, x_0) = \frac{(1-\bar{\alpha}_{t-1})\cdot\sqrt{\alpha_t}}{1-\bar{\alpha}_t}x_t + \frac{(1-\alpha_t)\cdot\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_t}\cdot x_0$$

$$\mu_\theta(x_t) = \frac{(1-\bar{\alpha}_{t-1})\cdot\sqrt{\alpha_t}}{1-\bar{\alpha}_t}x_t + \frac{(1-\alpha_t)\cdot\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_t}\cdot \hat{x}_\theta(x_t)$$

where $\hat{x}_\theta(x_t)$ is the output of a Neural Network.

with this, $\frac{1}{2\sigma_q^2}\|\mu_\theta - \mu_q\|_2^2$

p92 The network is a denoiser predicting x_0

$$= \frac{1}{2\sigma_q^2}\left\|\frac{(1-\alpha_t)\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_t}\left(\hat{x}_\theta(x_t) - x_0\right)\right\|_2^2$$

$$= \frac{1}{2\sigma_q^2}\cdot\frac{(1-\alpha_t)^2\bar{\alpha}_{t-1}}{(1-\bar{\alpha}_t)^2}\|\hat{x}_\theta(x_t) - x_0\|_2^2$$

> **DIAGRAM** x_t, t → [U-net (θ, decrease dim → increase dim)] → $\hat{x}_\theta(x_t)$ → $\|\hat{x}_\theta(x_t) - x_0\|_2^2$ (Regressor on the input data point x_0)

The above network can be viewed as a **"denoiser"** for the datapoint x_0, at all noisy instances x_t.

Unlike in a GAN/VAE, the Neural Network that gets trained in a DDPM, **does not output the novel data point from** P_{x_0} (true data dist).

p93 Training of a DDPM

Given a data point x_0 from the true data dist. **Repeat until convergence** ($\alpha_1 \dots \alpha_T$ are fixed $[0,1]$, $T \approx 1000$): 1. pick a $t \sim \text{uniform}[0,T]$ 2. $x_t^m \sim \mathcal{N}(x_t;\ \sqrt{\bar{\alpha}_t}x_0,\ 1-\bar{\alpha}_t I)$ $\quad : q(x_t|x_0)$

$$x_t^m = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\cdot\epsilon_t, \qquad \epsilon_t \sim \mathcal{N}(0,I)$$

3.

$$\theta^{k+1} \leftarrow \theta^k - \beta\,\nabla_\theta J_\theta(q)$$

$$\text{where } J_\theta(q) = \sum_{m=1}^{M} \|\hat{x}_\theta(x_t^m) - x_0\|_2^2, \qquad M \subset \{0,T\}$$

DIAGRAM x_t, t → [U-net θ] → $\hat{x}_\theta(x_t)$ → forward pass to compute $\hat{x}_\theta(x_t)$;

$$\nabla_\theta \sum_{m=1}^{M} \|\hat{x}_\theta(x_t^m) - x_0\|_2^2 \quad \text{Backward pass for Grad descent}$$

— End of training DDPM —

p94 Inference or Sampling in a DDPM

To get a sample $x_0 \sim P_{x_0}$, we need to traverse through the **backward / Decoding process**:

$$x_T \to x_{T-1} \to x_{T-2} \to \cdots \to x_0$$

$$\Rightarrow \text{ we need to sample iteratively from decoding distributions:}$$

$$x_{t-1} \sim P_\theta(x_{t-1} \mid x_t) \qquad \text{for } t = T \text{ to } 1$$

$$x_T \sim \mathcal{N}(0,I), \quad x_{T-1} \sim P_\theta(x_{T-1}|x_T), \quad x_{T-2} \sim P_\theta(x_{T-2}|x_{T-1}), \ \dots\ , \quad x_0 \sim P_\theta(x_0|x_1)$$

We have,

$$x_{t-1} \sim P_\theta(x_{t-1}|x_t)$$

$$P_\theta(x_{t-1} \mid x_t) = \mathcal{N}\big(x_{t-1};\ \mu_{\theta^*}(x_t),\ \sigma_q^2 \cdot I\big)$$

$$x_{t-1} = \mu_{\theta^*}(x_t) + \sigma_q \cdot \epsilon, \qquad \epsilon \sim \mathcal{N}(0,I)$$

$$\text{where } \mu_{\theta^*}(x_t) = \frac{(1-\bar{\alpha}_{t-1})\cdot\sqrt{\alpha_t}}{1-\bar{\alpha}_t}\,x_t + \frac{(1-\alpha_t)\cdot\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_t}\,\hat{x}_{\theta^*}(x_t)$$

p95 The inference algorithm

Inference in a DDPM: - start with $x_T \sim \mathcal{N}(0,I)$ - **for** $t = T, T-1, \dots, 1$ - **do:** - get $\hat{x}_{\theta^*}(x_t)$ from the NN

DIAGRAM x_t, t → [U-net θ^*] → $\hat{x}_{\theta^*}(x_t)$ forward pass to get $\hat{x}_{\theta^*}(x_t)$

-

$$x_{t-1} = \mu_{\theta^*}(x_t) + \sigma_q^2\epsilon, \qquad \epsilon \sim \mathcal{N}(0,I) \quad (\text{sampling from } P_\theta(x_{t-1}|x_t))$$

-

$$\mu_{\theta^*}(x_t) = \frac{(1-\bar{\alpha}_{t-1})\cdot\sqrt{\alpha_t}}{1-\bar{\alpha}_t}\, x_t + \frac{(1-\alpha_t)\cdot\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_t}\, \hat{x}_{\theta^*}(x_t)$$

- **return** x_0 $\therefore$ the new generated sample.

To sample in a DDPM, we need to perform **T number of forward passes** through the trained NN (θ^*).

— End of DDPM training & Inference —

p96 Alternate Interpretations of DDPM: (a) regression over noise

(a) DDPM as regression over added "noise"

Recall, the forward process of DDPM,

$$x_t = \sqrt{\bar{\alpha}_t}\cdot x_0 + \sqrt{1-\bar{\alpha}_t}\cdot\epsilon_t, \qquad \epsilon_t \sim \mathcal{N}(0, I)$$

Rearranging the above terms,

$$x_0 = \frac{x_t - \sqrt{1-\bar{\alpha}_t}\cdot\epsilon_t}{\sqrt{\bar{\alpha}_t}} \qquad -(1)$$

ϵ_t : noise added to x_0, to convert it into x_t.

Consistency term in the ELBO : $\|\mu_\theta - \mu_q\|_2^2$ - μ_q : mean of $q(x_{t-1}|x_t, x_0)$ - μ_θ : mean of $P_\theta(x_{t-1}|x_t)$

Represent μ_q, in terms of x_t & ϵ_t

$$\mu_q(x_t, x_0) = \frac{\sqrt{\alpha_t}\cdot(1-\bar{\alpha}_{t-1})x_t + \sqrt{\bar{\alpha}_{t-1}}\cdot(1-\alpha_t)x_0}{1-\bar{\alpha}_t}$$

Replacing x_0, above using Eq. 1

p97 μ_q and μ_θ in terms of the noise ε_t

$$\mu_q(x_t, x_0) = \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})\cdot x_t + \sqrt{\bar{\alpha}_{t-1}}\cdot(1-\alpha_t)\overbrace{\left(\frac{x_t - \sqrt{1-\bar{\alpha}_t}\,\epsilon_t}{\sqrt{\bar{\alpha}_t}}\right)}^{x_0}}{1-\bar{\alpha}_t}$$

$$\mu_q(x_t, \epsilon_t) = \frac{1}{\sqrt{\bar{\alpha}_t}}\cdot x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}\sqrt{\bar{\alpha}_t}}\cdot\epsilon_t$$

$$\mu_\theta(x_t) = \frac{1}{\sqrt{\alpha}}\cdot x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}\sqrt{\bar{\alpha}_t}}\cdot\hat{\epsilon}_\theta(x_t)$$

$$\text{consistency term} = \frac{1}{2\sigma_q^2}\|\mu_q - \mu_\theta\|_2^2$$

$$= \frac{1}{2\sigma_q^2}\cdot\frac{(1-\alpha_t)^2}{(1-\bar{\alpha}_t)\alpha_t}\left[\|\epsilon_t - \hat{\epsilon}_\theta(x_t)\|_2^2\right]$$

$$\propto \|\epsilon_t - \hat{\epsilon}_\theta(x_t)\|_2^2$$

Regression over the added noise.

DIAGRAM x_t, t → [U-net θ] → $\hat{\epsilon}_\theta(x_t)$ → $\|\epsilon_t - \hat{\epsilon}_\theta(x_t)\|_2^2$ (Regressor on the added noise)

p98 (b) DDPM as score-predictors

$$\text{suppose } t \sim \mathcal{N}(t; \mu_t, \Sigma_t) \doteq P(t)$$

Tweedie's formula:

$$\mathbb{E}\left[\mu_t \mid t\right] = t + \Sigma_t \cdot \nabla_t \log P(t)$$

$$\text{where } \nabla_t \log P(t) \text{ defined as the "score function"}$$

Recall, that in a DDPM,

$$q(x_t|x_0) = \mathcal{N}\left(x_t;\ \sqrt{\bar{\alpha}_t} \cdot x_0,\ (1-\bar{\alpha}_t)I\right)$$

Apply Tweedie's formula to above,

$$\mathbb{E}\left(\mu \mid x_t\right) = x_t + (1-\bar{\alpha}_t) \cdot \nabla_{x_t} \log P(x_t)$$

The best estimate for $\mathbb{E}(\mu|x_t)$ is the true mean. $\Rightarrow \mathbb{E}(\mu|x_t) = \sqrt{\bar{\alpha}_t} \cdot x_0$.

$$\sqrt{\bar{\alpha}_t} \cdot x_0 = x_t + (1-\bar{\alpha}_t) \cdot \nabla_{x_t} \log P(x_t)$$

Rearranging the terms,

$$\boxed{x_0 = \frac{x_t + (1-\bar{\alpha}_t) \cdot \nabla_{x_t} \log P(x_t)}{\sqrt{\bar{\alpha}_t}}}$$

p99 Consistency term in terms of score functions

Expressing the consistency term in terms of score-functions.

$$\mu_q(x_t, x_0) = \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1}) \cdot x_t + \sqrt{\bar{\alpha}_{t-1}} \cdot (1-\alpha_t) \overbrace{\left(\frac{x_t + (1-\bar{\alpha}_t)\nabla \log P(x_t)}{\sqrt{\bar{\alpha}_t}}\right)}^{x_0}}{1-\bar{\alpha}_t}$$

$$\boxed{\mu_q = \frac{1}{\sqrt{\bar{\alpha}_t}} \cdot x_t + \frac{1-\alpha_t}{\sqrt{\bar{\alpha}_t}} \cdot \nabla_{x_t} \log P(x_t)}$$

$$\mu_\theta = \frac{1}{\sqrt{\bar{\alpha}_t}} \cdot x_t + \frac{1-\alpha_t}{\sqrt{\bar{\alpha}_t}} \cdot S_\theta(x_t)$$

$S_\theta(x_t)$: predicted **score** using NN.

$$\text{consistency term} \propto \|\mu_q - \mu_\theta\|_2^2 \ \propto\ \|\nabla_{x_t} \log P(x_t) - S_\theta(x_t)\|_2^2$$

Regression over the score function.

DIAGRAM x_t, t → [U-net θ] → $S_\theta(x_t)$ → $\|\nabla_{x_t} \log P(x_t) - S_\theta(x_t)\|_2^2$ (Regressor on the score function)

p100 True score = negatively scaled noise

To compute the true score

$$x_0 = \frac{x_t + (1-\bar{\alpha}_t) \cdot \nabla_{x_t} \log P(x_t)}{\sqrt{\bar{\alpha}_t}} \qquad -(1)$$

$$\text{Also, } x_0 = \frac{x_t - \sqrt{1-\bar{\alpha}_t} \cdot \epsilon_t}{\sqrt{\bar{\alpha}_t}} \quad \text{by definition,} \qquad \epsilon_t \sim \mathcal{N}(0, I) \qquad -(2)$$

Equating (1) & (2), & rearranging

$$\nabla_{x_t} \log P(x_t) = \frac{-1}{\sqrt{1-\bar{\alpha}_t}} \epsilon_t$$

True score = negatively scaled noise

$\Rightarrow$ a DDPM trained to regress over added noise, is implicitly predicting the score function.

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p100 101 — Guided Diffusion Models for Conditional Generation

Data : (x_0, y) where y : conditioning variable Eg. x_0 : Image, \quad y : class-label, \quad y : text-Embedding

Goal: sample from the conditional distribution $P(x_0|y)$?

Question: How to modify a DDPM to sample from the cond. dist. $P(x_0|y)$?

Classifier Guidance

Recall that DDPMs are score-predictors, predicting the true score of the un-conditional or marginal distribution, $\nabla_{x_t} \log P(x_t)$.

From a conditional sampling perspective, the goal is to predict or regress over the **conditional score**, $\nabla_{x_t} \log P(x_t|y)$.

$$\nabla_{x_t} \log P(x_t|y) = \nabla \log \left(\frac{P(x_t) \cdot P(y|x_t)}{P(y)} \right)$$

$$= \nabla_{x_t} \log P(x_t) + \nabla_{x_t} \log P(y|x_t) - \nabla_{x_t} \log P(y)$$

$$\nabla_{x_t} \log P(x_t|y) = \nabla_{x_t} \log P(x_t) + \nabla_{x_t} \log P(y|x_t)$$

p102 Classifier guidance in practice; Classifier-Free Guidance

DIAGRAM y (), x_t, t → [U-net θ] → $\|\varepsilon_t - \hat{\varepsilon}_\theta\|_2^2$ (predict unconditional score) → $\hat{\varepsilon}_\theta(\nabla_{x_t} \log p(x_t))$

$$\nabla_\theta \|\epsilon_t - \hat{\epsilon}_\theta\|_2^2 + \nabla_{x_t} \log P(y|x_t)$$

$$x_t \rightarrow \boxed{\text{classifier / Regression (pre-trained)}} \rightarrow P(y|x_t) \rightarrow \nabla_{x_t} \log P(y|x_t)$$

top labels: "unconditional score" (over the U-net output), "Classifier gradient"

classifier / Regressor is trained **separately** to predict y (class label or Embedding) from x_t. **Training the classifier at different noise levels is difficult.**

Classifier-Free Guidance

$$\nabla \log P(x_t|y) = \nabla \log P(x_t) + \nabla \log P(y|x_t)$$

$$\nabla \log P(y|x_t) = \nabla \log P(x_t|y) - \nabla \log P(x_t) \qquad \text{(Baye's law)}$$

p103 The classifier-free guidance formula

$$\nabla \log P(x_t|y) \approx \nabla \log P(x_t) + \lambda \Big(\nabla \log P(x_t|y) - \nabla \log P(x_t) \Big)$$

$$\boxed{\nabla_{x_t} \log P(x_t|y) = \lambda \nabla_{x_t} \log P(x_t|y) + (1-\lambda) \nabla_{x_t} \log P(x_t)}$$

λ : hyper-parameter.

> **DIAGRAM** y, x_t, t → [U-net θ] → ⋯ (Regressor over ∇_$x_t$log P(x_t|y))

$$\text{null label / text } y = \phi: \quad x_t,\ t \to \boxed{\text{U-net } \theta \text{ (copy of the above)}} \to \cdots \quad (\text{Regressor over } \nabla_{x_t} \log P(x_t))$$

add both the (λ) cond. & $(1-\lambda)$ uncond. scores before the gradient step.

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p104 Latent Diffusion Models

The basic idea: Build DMs on the **latent space** induced by another Encoder-Decoder Models.

Given data $x_0 \in \mathbb{R}^d \sim p(x_0)$, the first step is to learn a latent representation $z_0 \in \mathbb{R}^k$, $k \ll d$, using an Enc-Dec Model.

Eg: construct a **Vq-VAE** to learn a latent space.

> **DIAGRAM** x_0 → [Enc] → z_0 → [Dec] → $\hat{x_0}$

Let E_{ϕ^*}, D_θ represent the Encoder & Decoder functions, respectively. E_{ϕ^*} & D_θ are trained using the original data x_0 or some other dataset similar to x_0.

All the datapoints in $\mathbb{R}^d$, are projected on to the latent space as

$$z_0 = E_{\phi^*}(x_0)$$

p105 Latent DM inference; DDIMs motivation

A **DDPM is constructed on the space of** z_0. During inference, to generate a novel point from $p(x_0)$, first sample a novel point from the latent space with reverse diffusion, z_{novel} & $x_{novel} = D_{\theta^*}(z_{novel}) \sim p(x_0)$.

Denoising Diffusion Implicit Models (DDIMs)

- DDPMs are **slow in sampling** ($T = 1000$) & cannot be used for **deterministic posterior Inference**.
- The forward process is **stochastic**.

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon_t, \qquad \epsilon_t \sim \mathcal{N}(0, I)$$

$$\Rightarrow \text{ Given an } x_0, \text{ running the forward process multiple times leads to different } x_T$$

(final latent representation for x_0)

$$\Rightarrow \text{ There does not exist a single / unique } x_T$$

p106 DDIM formulation

...that generates a given x_0, from the reverse diffusion process.

The above issues (**slow-sampling** & **non-unique invertibility**) of DDPMs are solved with **DDIMs**.

DDIMs

Let us consider a family of inference distributions, parameterized by a vector $\sigma \in \mathbb{R}_{\geq 0}$

$$q_\sigma(x_{1:T} \mid x_0) \doteq q_\sigma(x_T|x_0) \prod_{t=2}^{T} q_\sigma(x_{t-1} \mid x_t, x_0)$$

$$\text{where } q_\sigma(x_T|x_0) \doteq \mathcal{N}\big(\sqrt{\bar{\alpha}_T}\, x_0,\ (1-\alpha_T)I\big)$$

and for all $t > 1$

$$q_\sigma(x_{t-1}|x_t, x_0) \doteq \mathcal{N}\left(\underbrace{\sqrt{\bar{\alpha}_{t-1}}\, x_0 + \sqrt{1-\alpha_{t-1}-\sigma_t^2} \cdot \frac{x_t - \sqrt{\bar{\alpha}_t}\, x_0}{\sqrt{1-\bar{\alpha}_t}}}_{\text{mean}},\ \sigma_t^2 I\right)$$

The forward process corr. to the above posterior is:

$$q_\sigma(x_t \mid x_{t-1}, x_0) = \frac{q_\sigma(x_{t-1}|x_t, x_0) \cdot q_\sigma(x_t|x_0)}{q_\sigma(x_{t-1}|x_0)} \qquad \text{(Baye's rule)}$$

p107 Why DDIM and DDPM share the same ELBO

The above forward process is **not Markovian**.

$$\Rightarrow\ x_t \not\perp\!\!\!\perp x_0 \mid x_{t-1}$$

$$\boxed{\text{In the above process, } q_\sigma(x_t|x_0) = \mathcal{N}\big(x_t;\ \sqrt{\bar{\alpha}_t} \cdot x_0,\ \sqrt{1-\alpha_t} \cdot I\big)}$$

the same as DDPM.

Since $q_\sigma(x_t|x_0)$, conditional of x_t given x_0 is the **same in both DDPM & DDIM**, the ELBO for both of them are the same & optimizing the ELBO leads to the same solution.

$\therefore$ the ELBO in DDPM / DDIM depends **only on** $q(x_t|x_0)$

$$\Rightarrow\ \text{as long as } q(x_t|x_0) \text{ is same as that of DDPMs, any process can be used}$$

& leads to the same solution for ELBO. as far as $q(x_t|x_0)$ is same as DDPM, any definition for $q(x_{1:T}|x_0)$ can be chosen.

p108 Defining the Decoder / Reverse process

$$P_\theta(x_{t-1} \mid x_t) = \begin{cases} \mathcal{N}\big(\mu_\theta(x_1),\ \sigma^2 I\big) & t = 1 \\ q_\sigma(x_{t-1} \mid x_t,\ \mu_\theta(x_t)) & \text{otherwise} \end{cases}$$

$$\text{where } \mu_\theta(x_t) = \frac{x_t - \sqrt{1-\alpha_t} \cdot \hat{\epsilon}_\theta}{\sqrt{\alpha_t}} \qquad : \text{ approx for } x_0$$

Recall,

$$x_t = \sqrt{\alpha_t} \cdot x_0 + \sqrt{1-\alpha_t} \cdot \epsilon_t, \qquad \epsilon_t \sim \mathcal{N}(0, I)$$

$$x_0 = \frac{x_t - \sqrt{1-\alpha_t} \cdot \epsilon_t}{\sqrt{\alpha_t}}$$

The main result: The ELBO corresponding to the above non-markovian family of models & that of DDPM are the **same** (off-by a const.)

$$\boxed{\Rightarrow\ \text{Training the regression network in DDPM is implicitly training all the above non-Markovian Models.}}$$

There is **no difference** between training DDPM & DDIMs, but the **inference process is different**.

p109 Inference with DDIMs

Sampling from $P_\theta(x_{t-1}|x_t)$

$$P_{\theta^*}(x_{t-1} \mid x_t) = \begin{cases} \mathcal{N}\big(\mu_{\theta^*}(x_1),\ \sigma^2 I\big) & t = 1 \\ q_\sigma(x_{t-1} \mid x_t,\ \mu_{\theta^*}(x_t)) & \text{otherwise} \end{cases}$$

$$\mu_\theta(x_t) = \frac{x_t - \sqrt{1-\alpha_t}\cdot\hat{\epsilon}_\theta}{\sqrt{\alpha_t}} \qquad \Longrightarrow \qquad \frac{x_t - \sqrt{\bar{\alpha}_t}\,\mu_{\theta^*}}{\sqrt{1-\bar{\alpha}_t}} = \hat{\epsilon}_{\theta^*}$$

$$q_\sigma(x_{t-1} \mid x_t, \mu_{\theta^*}) \doteq \mathcal{N}\left(\sqrt{\bar{\alpha}_{t-1}}\,\mu_{\theta^*} + \sqrt{1-\alpha_{t-1}-\sigma_t^2}\cdot\frac{x_t - \sqrt{\bar{\alpha}_t}\,\mu_{\theta^*}}{\sqrt{1-\bar{\alpha}_t}},\ \sigma_t^2 I\right)$$

putting all the above together, sampling from $P_{\theta^*}(x_{t-1}|x_t)$ is as follows: $z \sim \mathcal{N}(0, I)$

$$\boxed{x_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\left[\frac{x_t - \sqrt{1-\alpha_t}\cdot\hat{\epsilon}_{\theta^*}}{\sqrt{\alpha_t}}\right] + \sqrt{1-\alpha_{t-1}-\sigma_t^2}\,\hat{\epsilon}_{\theta^*} + \sigma_t^2\cdot z}$$

p110 DDPM as a special case; DDIM = deterministic

when

$$\sigma_t = \frac{\sqrt{1-\alpha_{t-1}}}{\sqrt{1-\alpha_t}}\cdot\frac{\sqrt{1-\alpha_t}}{\sqrt{\alpha_{t-1}}}$$

the above random process reduces to **DDPM**.

when $\sigma_t = 0$, the sampling process has the following form:

$$\boxed{x_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\left[\frac{x_t - \sqrt{1-\alpha_t}\cdot\hat{\epsilon}_{\theta^*}}{\sqrt{\alpha_t}}\right] + \sqrt{1-\alpha_{t-1}}\,\hat{\epsilon}_{\theta^*}}$$

- The above sampling process is **deterministic**.
- The only randomness is from $x_T \sim \mathcal{N}(0, I)$
- with $\sigma_t = 0$, the above model is called **DDIM**.

$$\Rightarrow \text{ the forward process in DDIM with } \sigma_t = 0, \text{ is also deterministic.}$$

- The above implies that DDIM has a **unique invertibility**. (**DDIM Inversion**)

— End of Diffusion Models —

Transcribed page-by-page from *GenAI-IITM.pdf* (handwritten notes, pages 72–110). Notation preserved as written. Built for revision — verify against the source for coursework.